Statement of Purpose Amazon ML Summer School

Honestly, I didn't come to machine learning the way most people describe it. There was no single "aha moment," no inspiring lecture, no mentor who pointed me in this direction. I came to it through back pain. Actual, annoying, persistent back pain from sitting hunched over a laptop for hours during my first year of college. I kept telling myself I'd fix my posture eventually. I never did , until I thought why not just build something that fixes it for me? That project, my Real-Time Posture Detection System, ended up teaching me far more than I expected. I used TensorFlow MoveNet and OpenCV to detect body keypoints in real time, computed alignment angles across the shoulder, hip and knee, and triggered desktop alerts whenever posture slipped. I added email reports and session logging with Pandas so users could track themselves over time, not just in the moment.

But the part that really stuck with me wasn't the model. It was everything after. Weeks of false alerts, threshold tuning, edge cases I hadn't thought of. I learned that getting a model to work and getting a system to reliably work are two very different problems. That question has followed me into every project since. After that, I started contributing to open source, partly to learn and partly because I wanted to see how other people solve problems I hadn't encountered yet. I raised issues, improved documentation, and contributed to both ML and general development repositories

I also participated in a hackathon, where I had to make real architecture decisions under time pressure. I found that I think differently when something actually has to work by a deadline. Here's the part I want to be honest about.

I study at a tier-3 college in Andhra Pradesh. My professors are genuinely supportive, but there is nobody in my immediate environment who has built a recommendation system for millions of users, optimized a deep learning model for production, or designed an NLP pipeline that handles real-world noise at scale. Almost everything I know has come from YouTube, documentation, research papers, and a lot of trial and error. I don't say this as a complaint.

I'm proud of how far I've come without a structured path. **But I'm also clear-eyed about what I'm missing and that is depth that only comes from people who have actually done this**. The Amazon ML Summer School curriculum, specifically the tracks on Deep Learning, Natural Language Processing, Generative AI, and Large Scale Machine Learning, covers exactly the gaps I haven't been able to fill on my own. Learning these directly from Amazon scientists is something I genuinely cannot replicate from where I am. My long term goal , is to work on AI in healthcare specifically, building tools that can detect early signs of disease from medical imaging, the kind of quiet, reliable technology that helps doctors catch what human eyes might miss.

I believe this program is the most direct path to that. I'm deeply grateful to the Amazon ML Summer School team for creating a program that gives students like me — self-taught, from smaller colleges, far from the usual networks — a real chance to learn from the best.

Thank you for this opportunity,

Pujitha Bolisetty